TSANG Cheuk Nam

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Summary

Final-year HKUST Industrial Engineering & Engineering Management student (Extended Major in AI) who redesigns traditional processes to be AI-native and builds the AI systems behind them, directing AI coding agents (Claude Code, Codex) that write about 95% of the code. Won a hackathon track by reinventing insurance this way. Internship now; full-time from January 2027.

Education

The Hong Kong University of Science and Technology (HKUST) Hong Kong
BEng in Industrial Engineering & Engineering Management; Extended Major in AI *Sep 2022 – Dec 2026 (expected)*

Honors: Winner, "Reinventing Insurance for an AI-native World" track (sponsored by bolttech), Hong Kong AI Summit: BUILD ACROSS Hackathon, 200+ participants (Oct 2026)

Coursework: Statistical Machine Learning; Generative AI and LLMs for Operations Research; Stochastic Models

Projects

Budget-Based Microinsurance | *Winning concept, built in four days* · Rust, TypeScript, Python Oct 2026

- Designed an original AI-native insurance model: customers pick the price and AI builds the cover, answering all 21 pain points mapped across insurers, partners and customers.
- Designed 9 AI components to take routine work off agents and claims handlers, so cover starts at HK$40 a month.
- Made every AI component fail safe: checked structured answers, saved fallbacks, and 36 evals including deliberately bad replies.
- Designed where AI stops: it recommends, fixed rules set every price and payout, and only a person can refuse a claim.
- Settles automatic payouts for 1 million policies in 1.6 seconds, against a 5-second target.

LLM Brand Recommendation Benchmark | *AI-native market research* · Python, SQLite, LLM APIs Sep 2026

- Turned a manual brand-research task into an AI-run benchmark of which brands AI assistants recommend in each product category.
- Redesigned the method after the first approach passed one category; 50 of 51 categories then passed validation against verified brand lists within two days.
- Traced most failures to incomplete reference data, not the model; fixing the data turned 13 of 14 into passes.
- Cut evaluation cost by two thirds and kept results honest: AI answers count only as leads until independently verified.

Using LLMs for Optimization | *IEDA Final Year Project — Experiment & Benchmarking Lead* Completed 2026

- Showed an LLM works best as a planner inside a checked workflow: letting it plan instead of writing the whole program cut AI cost 83% at the same quality.

Self-Hosted Personal AI Agent | *Always-on AI agent and coding-agent team* · Python, Kotlin Aug 2026 – Present

- Runs a self-hosted personal AI agent 24/7 and a team of coding agents, with eight rules, each from a real failure, that stop agents building what nobody asked for.

Polymarket Market-Making & Forecasting Models | *AI-agent-driven quantitative research* Aug 2026 – Present

- Built football forecasting models that beat the standard Elo rating on 3,504 held-out matches, then moved the research to market making.
- Replayed 777,588 orders to size risk before quoting: prices hit sell quotes about 4.5× as often as buy quotes.

9UPPER Online | *Multilingual real-time party game* · Next.js, TypeScript, Supabase Mar 2026 – Present

- Built a multilingual real-time game platform with AI coding agents and grew it through SEO, GEO and AEO to 1,789 Google clicks in three months; ChatGPT and Claude search now recommend it.

Skills

AI application design: LLM components, evals and guardrails, structured outputs, model routing, human-in-the-loop

AI-native transformation: process redesign, deciding what AI, rules or people handle, cost and quality measurement

Agentic development: Claude Code, Codex, multi-agent orchestration, harness and context engineering

Software: Python, TypeScript, React, Rust, PostgreSQL, REST APIs, Docker, Linux, Git

Languages: Cantonese (native), English (fluent), Mandarin (fluent)